VI Empirical Portfolio Learnability

A Data, Representation, and Investment Performance

The empirical exercise asks how a representation of stock characteristics, finite return history, and regularization jointly shape a portfolio policy. A richer representation makes additional portfolio opportunities available, but a finite return history does not necessarily make all those opportunities learnable or financially valuable. We first compare Linear, Gaussian, and Matérn-3/2 policies on identical subsequent months. We then fix the Gaussian representation to examine regularization and the economic content of its additional directions. The evidence is descriptive historical evidence: neither a favorable Sharpe nor an estimated number of active directions certifies a reliably profitable opportunity.

We use the monthly U.S. equity characteristic panel of Jensen, Kelly, and Pedersen, with formations from January 1963 through December 2024. Starting from their research dictionary, we exclude nano stocks, retain the 130 characteristics with greatest coverage in 1963–1972, and remove observations missing more than 30% of these inputs. Observed values receive average cross-sectional ranks on $[-0.5, 0.5]$; remaining missing inputs receive zero. The panel is unbalanced. The retrospectively constructed characteristic dictionary and current data snapshot constitute the academic information set, rather than a reconstruction of historical signal discovery or data vintages.

There is a material sample limitation. Unavailable subsequent payoffs are excluded before ranking and calculating the number of stocks N_t. A formation-only audit leaves 7,489 unidentified stock-month payoffs out of 2,504,402 eligible observations (0.299%); documented adjacent-calendar source matches recover none. Formation-only ranks are invariant to perturbing future payoff availability, whereas the retained complete-payoff sample is conditional on that availability. Thus selection uses only earlier observed payoff values, but sample formation still depends on whether future payoffs are available. Unknown returns are never assigned zero. All performance and cost results retain this caveat.

The first inner training period uses 1963–1972 formations and validation uses 1973–1977. Each subsequent January-close decision reserves the preceding 60 formation months for validation, then refits on the full available pretest history. December's payoff is known at January close. January–December formation positions therefore realize February–next January returns. All three policies are evaluated on the same 564 OOS months, February 1978–January 2025.

Linear uses a constant and the characteristic coordinates. Gaussian and Matérn-3/2 use 10,000 random Fourier features with seed 0. For each kernel, chronological validation jointly selects lengthscale and ridge penalty by minimizing the mean response-one quadratic loss Q at fitted scale. Lengthscales are $\{0.25, 0.5, 1, 2, 4\}$ times the initial-sample median distance; each has 120 penalties fixed from its own initial training spectrum. Frequency and phase draws are shared across lengthscales. The chosen pair is refitted using training and validation observations before the test year; no test payoff selects either parameter. Linear selects its penalty by the same chronological loss criterion. This is a comparison of representations under past validation; finite feature banks do not establish strict nesting or identify a population approximation gap from an observed performance ranking.

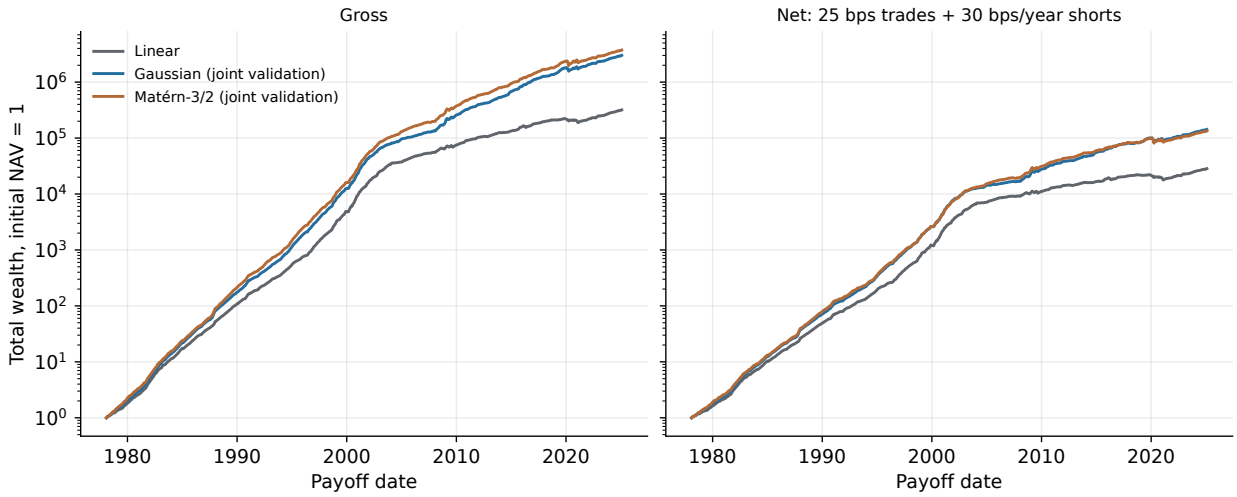


Figure 3: Investment performance on identical OOS months. Linear validates its ridge penalty; Gaussian and Matérn-3/2 jointly validate lengthscale and penalty using earlier observations. Total wealth compounds cash returns plus portfolio excess payoffs, starting from NAV one at January 1978 close. The right panel applies 25-basis-point commissions to both purchases and sales against drift-adjusted holdings, plus a 30-basis-point annual short fee. These are illustrative costs on a sample conditional on future-payoff availability.

Table I: Representation comparison under chronological validation

Policy	Mean %	Vol. %	SR_G	SR_N	Gross	Trades	Fees %
Linear	23.44	7.58	3.091	2.476	2.01	19.45	5.14
Gaussian	28.34	7.77	3.646	2.857	2.44	24.53	6.48
Matérn-3/2	28.85	8.04	3.587	2.747	2.58	26.82	7.07

All policies use the same 564 months, February 1978–January 2025, and common initial-validation exposure scale. Gaussian and Matérn jointly validate lengthscale and penalty; Linear validates penalty. Mean and volatility are annualized gross excess returns. Net Sharpe applies 25 bps two-sided commissions and 30 bps/year short fees. Gross is mean account gross/NAV; trades are annual drift-adjusted traded notional/NAV on the net path; fees are annual trading plus short charges/NAV. Costs are illustrative assumptions.

A common positive scale, 0.0374000, maps the initial Linear validation policy's median gross exposure to 1.8. It is fixed before the first test payoff and applies to all policies, without future volatility or full-sample exposure normalization. It leaves gross Sharpe unchanged but matters for costs. The Gaussian-minus-Linear gross Sharpe difference is 0.554, with paired 12-month-block interval [0.275,0.880]; the net difference is 0.381, with interval [0.140,0.653]. The corresponding Matérn differences are 0.496 [0.133,0.907] gross and 0.271 [-0.026,0.611] net. These comparisons condition on the fitted paths and use the same resampled months for each policy. Trading requirements and uncertainty accompany the point estimates; the comparison does not establish uniform dominance of one representation.

B A Fixed Economic Spectrum and Active Complexity

For the remainder of the section we use the existing fixed-lengthscale Gaussian specification: initial median distance 4.392108, 10,000 seed-0 features, and the original past-validation ridge selections.

Its full-policy gross/net Sharpes are 3.300/2.605. These are distinct from the jointly validated Gaussian returns in Figure 3 and Table I. Fixing lengthscale and feature draws keeps the opportunity representation unchanged: varying λ then changes regularization within the same managed-payoff operator. A lengthscale change would alter both that operator and the portfolios being decomposed.

Let G contain historical monthly managed payoffs $X_t = N_t^{-1} \sum_i R^e_{i,t+1} \phi(Z_{i,t})$. Response-one ridge fits $a_\lambda = (G'G/T + \lambda I)^{-1} G'\mathbf{1}/T$ for positive λ. The uncentered second-moment spectrum of $G'G/T$ orders directions by historical payoff variation. An eigenvalue is neither invested capital, expected return, Sharpe, nor a uniquely named anomaly. Figure 4 uses only the history available at January 2024 close. Raw eigenvalues preserve representation units; division by the trace is a concentration diagnostic that does not rescale the fitted policy.

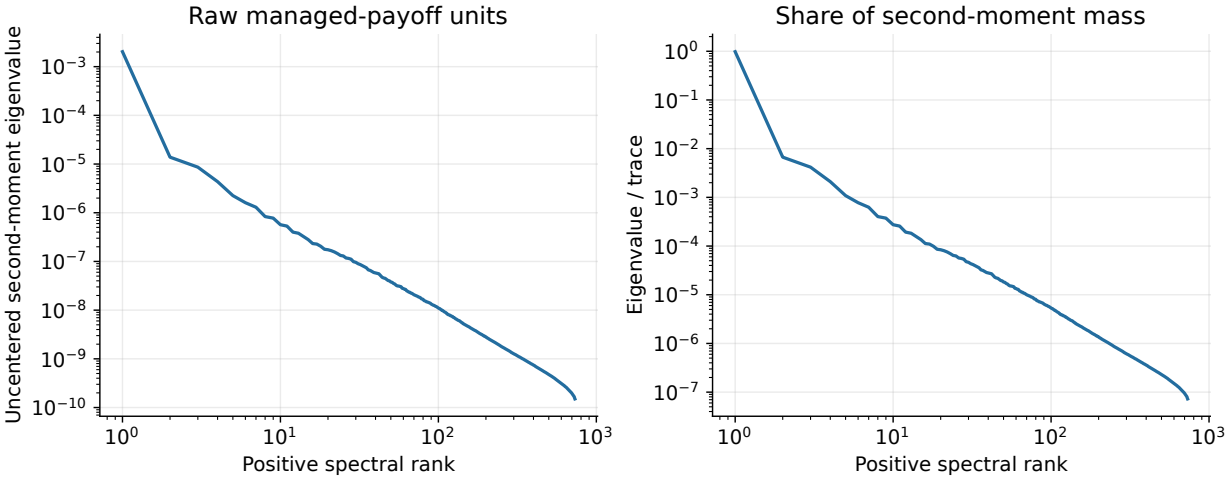


Figure 4: Managed-payoff spectrum of the fixed Gaussian representation at January 2024 close. Formations through December 2023 give 732 known monthly payoffs. The left panel shows raw uncentered second-moment eigenvalues; the right divides by their trace. Both use the same fixed initial median lengthscale and feature bank as Figures 5–7 and the spectral accounts in Table III.

Mass concentration and active complexity are distinct. Ridge weights a direction by $\widehat{\mu}_j/(\widehat{\mu}_j + \lambda)$, giving $C(\lambda) = \sum_j \widehat{\mu}_j/(\widehat{\mu}_j + \lambda)$. Weak eigenvalues can contribute many effective degrees of freedom while carrying little second-moment mass. Across annual histories, the first 10% of positive Gaussian ranks contain 99.921% of mass, but contribute only 33.8 of the selected policy's 124.5 mean effective directions. Positive rank uses a relative 10^{-12} eigenvalue tolerance; near-tied eigenvalues are kept together at group boundaries. Feasible complexity cannot exceed this historical effective rank, which is at most the number of historical monthly observations. A long weak tail can change exposures without establishing that its opportunities are learnable.

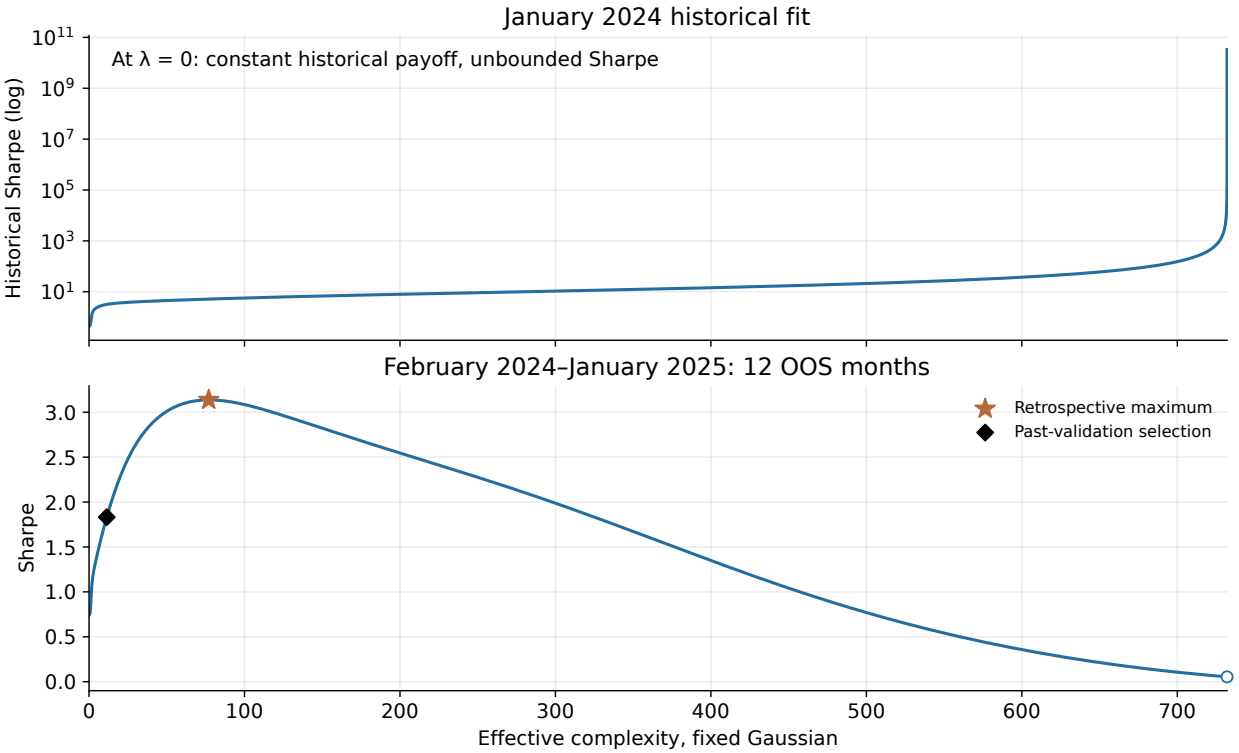


Figure 5: Complete regularization path for the fixed Gaussian January 2024 refit. Historical Sharpe uses a logarithmic scale; subsequent Sharpe uses a separate linear scale. Exact response-one interpolation has zero historical variance and unbounded historical Sharpe; unresolved numerical ratios near that limit are omitted from the historical panel only. Each positive-penalty point is evaluated on its actual February 2024–January 2025 OOS payoffs. The open circle is the exact minimum-norm zero-penalty policy at effective rank 732. The star is the retrospective maximum over the extended evaluation grid; the diamond retains the original earlier-validation selection. Neither is an estimate of a significant population optimum.

The final-year path holds lengthscale, feature bank, and history fixed. Its retrospective maximum is at complexity 77.0 with subsequent Sharpe 3.138; at zero penalty Sharpe is 0.054. Historical fit and subsequent investment performance need not move together. Twelve subsequent months produce a noisy illustration, so Figure 6 provides the complete progression in nine historical blocks.

The evaluation grid retains every original penalty and adds a logarithmic sequence from strong regularization to very small positive penalties, followed by exactly $\lambda = 0$. Its 1,522 policies are evaluated separately at every annual refit. The zero endpoint uses a symmetric eigensystem Moore–Penrose pseudoinverse with the same positive-rank tolerance, rather than an inverse of a singular operator. Direct feature-space least squares checks the first and final endpoints, and the smallest positive penalties converge numerically to the zero returns. No added evaluation changes the original validation policy, and no OOS result selects a deployable penalty.

To assess whether the final-year Sharpe–complexity profile is representative of the broader historical record, Figure 6 repeats the analysis across nine nonoverlapping investment periods. Holding the Gaussian representation fixed, we vary only the degree of regularization and evaluate each candidate on subsequent portfolio returns. The profiles show that realized performance need not improve monotonically as additional portfolio directions are activated. In several periods, Sharpe

reaches an interior maximum and subsequently declines, although the magnitude of the decline varies considerably across investment histories.

The annual cross-validation portfolio remains an implementable, past-selected policy, but it is not a single point on the block-level fixed-penalty path because the selected penalty can change from one annual refit to the next.

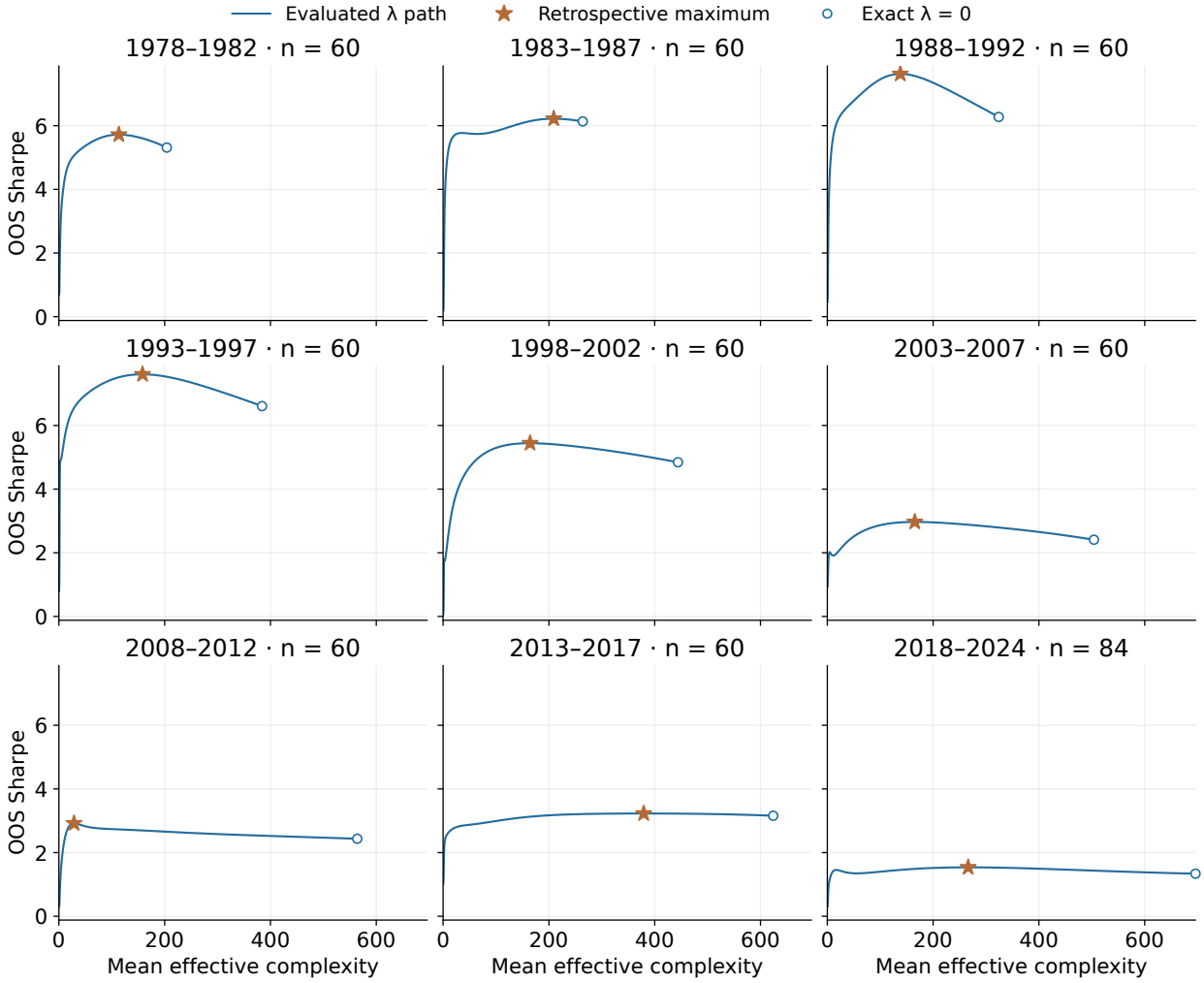


Figure 6: Historical out-of-sample Sharpe–complexity profiles. Each panel reports the realized out-of-sample Sharpe ratio associated with the complete ridge regularization path of the fixed-lengthscale Gaussian representation. The nine panels cover eight five-year decision blocks and one 2018–2024 block. For each candidate penalty, monthly out-of-sample returns are concatenated across the annual refits in the block, and a single Sharpe ratio is computed. Effective complexity is averaged across those refits, weighted by the number of evaluated months. Curves connect the actual evaluated penalties, including the extended positive grid and the exact minimum-norm zero-penalty endpoint, without smoothing or extrapolation. Stars indicate retrospective test-sample maxima; open circles indicate the unregularized endpoints. The dynamically selected annual cross-validation strategy is not plotted as a point on these curves because its penalty may differ across annual refits. All panels use common axes. The historical maxima are descriptive and are not estimates of population-optimal complexity.

Table II: Retrospective maxima and exact unregularized endpoints

Decision years	C at max.	Max. SR	$C(0)$	SR at $\lambda = 0$	Decline
1978–1982	113.4	5.719	204	5.313	0.406
1983–1987	209.1	6.219	264	6.134	0.085
1988–1992	137.9	7.626	324	6.274	1.352
1993–1997	158.2	7.608	384	6.609	0.999
1998–2002	164.4	5.443	444	4.844	0.599
2003–2007	165.4	2.970	504	2.410	0.559
2008–2012	28.7	2.918	564	2.435	0.483
2013–2017	379.2	3.228	624	3.158	0.070
2018–2024	266.2	1.537	696	1.337	0.200

All maxima use the entire extended grid. Complexity is the realized-month-weighted mean across annual refits; $C(0)$ equals mean historical effective rank. Decline is maximum minus zero-penalty Sharpe, a descriptive change rather than a significance test. The smallest-decline period is retained as observed.

All nine retrospective maxima are interior on the extended grid. Maximizing mean complexity ranges from 28.7 to 379.2. The 2008–2012 maximum occurs at low complexity. The unregularized endpoints remain close to the maxima in 1983–1987 and 2013–2017, with peak-to-zero Sharpe declines of 0.085 and 0.070, respectively. Neither period displays a substantial endpoint deterioration. The 2018–2024 decline is also modest; we do not impose a common downturn. Across blocks, the endpoint decline ranges from 0.070 to 1.352.These are descriptive shapes, not evidence of a statistically identified optimum in every period. Broad regions can be nearly flat relative to sampling uncertainty. All displayed maxima have been recomputed on the extended grid, including zero.

Intervals use 5,000 paired circular-block resamples, sharing monthly draws across all candidates or contrasted portfolios. They condition on estimated policies, representations, and holdings; they do not repeat model selection. Approximate stability and dependence decay within a block motivate this diagnostic. It neither corrects for searching the grid nor establishes validity under arbitrary nonstationarity. Six- and 24-month-block results for the representation comparisons and economic contrasts accompany the canonical results. A favorable retrospective peak is consequently distinct from a learnable policy selected using past information.

C What Additional Directions Trade

We interpret portfolios associated with eigenspaces of the same fixed Gaussian policy. At each annual refit, four disjoint positive-rank groups partition its ridge coefficients. Their actual contributions are reconstructed as stock-level signed holdings, providing long and short characteristic profiles and subsequent payoff and covariance attribution. The same selected penalty is retained across nested comparisons. These are attribution portfolios within one policy, not independently optimized strategies. Near-tied eigenvectors can rotate, so group profiles carry the historical interpretation rather than persistent names assigned to individual eigenvectors.

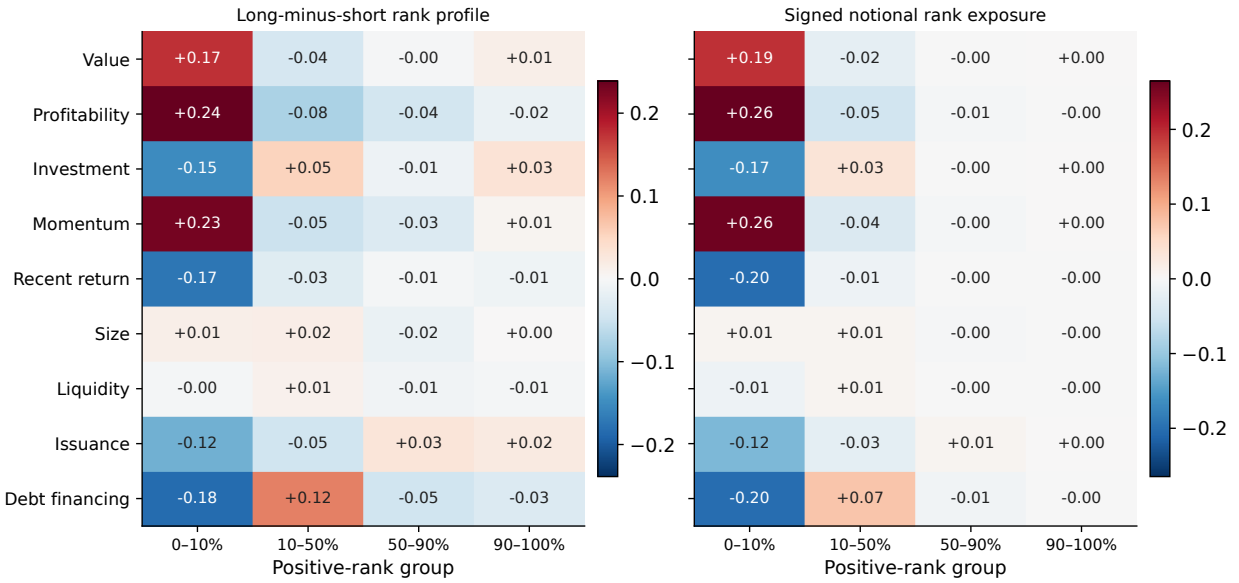


Figure 7: Economic profiles of the fixed Gaussian policy's actual spectral holdings. The left panel gives long-minus-short weighted formation-rank scores; the right preserves investment scale through signed notional exposures. An explicit signed characteristic dictionary defines the families. Debt financing and equity issuance are distinct. Families can overlap and cannot be added as disjoint return components. These holdings decompose the fixed representation, independently of the tuned-kernel opening comparison.

Leading holdings favor profitability and oppose recent returns and debt financing. Intermediate directions modify those exposures within the same characteristic universe. A large normalized characteristic contrast can coexist with small notional exposure, so profiles and signed notionals answer different questions. Formation descriptors characterize what is held; subsequent outcomes determine its investment contribution.

Table III: Fixed Gaussian spectral groups and incremental investment value

Ranks	Mass %	Own gross	Account gross	Mean %	SR_G	SR_N
0–10%	99.9215	2.267	2.267	24.323	3.012	2.354
10–50%	0.0685	1.263	2.333	2.759	3.284	2.598
50–90%	0.0091	0.327	2.325	0.047	3.286	2.594
90–100%	0.0010	0.078	2.326	0.026	3.300	2.605

Added ranks	ΔQ	Cross term	Own square	Paired 95% interval
0–10%	-0.402340	-1.083912	0.681573	[-0.524808, -0.257425]
10–50%	-0.036008	-0.131766	0.095758	[-0.093264, +0.027912]
50–90%	+0.000168	-0.003978	0.004146	[-0.008734, +0.009000]
90–100%	-0.003123	-0.003547	0.000424	[-0.006796, -0.000185]

The original selected penalty is retained at each annual refit. Mass averages annual historical second-moment shares. Own gross and mean describe individual components; account gross and Sharpes describe all groups through that boundary. The lower panel reports raw fitted-scale loss changes, relative to the null portfolio for the first group, with the same 5,000 paired 12-month-block draws. Weights are summed at stock level before trades and short fees. Gross Sharpe increments are +0.272, +0.002, +0.014; net increments are +0.244, -0.004, +0.011.

The leading group has mean component gross 2.267 and annual excess payoff 24.32%. Adding intermediate directions contributes 2.76% annually. Their own gross 1.263 increases the cumulative account's gross by only 0.066: some added positions offset existing stock holdings. Component gross exposures therefore do not add to the combined account's gross. The component's annual covariance with the leading portfolio is -0.000634, while total annual variance changes by 0.000278. Negative covariance can coexist with increased account variance because the component also adds its own variance.

Gross Sharpe rises from 3.012 to 3.284, and net Sharpe from 2.354 to 2.598. Paired 12-month-block intervals for these changes are [0.002,0.546] and [0.011,0.487]. The response-one loss changes by -0.036008, with interval [-0.093264,0.027912]; its cross term is -0.131766 and own-square term 0.095758. Thus a positive standalone mean, a Sharpe increment, and an incremental Q improvement are distinct assessments. The incremental identity $\Delta Q = -2\,\overline{(1-P)A} + \overline{A^2}$ includes the interaction of the previous policy P and added payoff A, at their original fitted scale.

Small incremental account gross does not imply small turnover. Monthly characteristics and annual refits can change offsetting positions substantially. For every nested account, stock weights are summed before calculating trades against drift-adjusted holdings, short notional, and net returns. Costs begin with entry from cash. Target weights refer to post-execution NAV; proportional commissions are solved jointly with the target cash residual. The annual short fee is divided by twelve. Cash earns or pays the same monthly risk-free rate, and terminal holdings are marked rather than forcibly liquidated. Separate component fees would not reconstruct the netted account. Changes in target weights remain a separate turnover proxy, never observed execution costs.

Corporate-action and delisting treatment relies on the upstream JKP return construction and stable identifiers; there is no independent share-count or execution-tape reconstruction. Liquidated names incur trades but receive no invented next payoff. The cost assumptions omit market impact and additional funding spreads. They are self-financing friction sensitivities, with the same future-payoff-availability caveat as gross returns.

The weak tail has little historical second-moment mass and small absolute payoff contributions. The successive gross/net Sharpe increments in Table III include adverse as well as favorable outcomes; the loss decomposition also retains the own-square and cross terms. Small mass alone cannot rule out covariance value, while a striking normalized characteristic tilt need not have a material financial contribution. Six-benchmark regressions of the actual component returns give descriptive spanning statistics; a separate hedge uses only pretest slope estimates. Neither supplies an independently selected investment strategy or a newly identified risk premium.

Economic content and supported complexity vary with history. The stationary theory is a benchmark for approximately stable conditions; a local interpretation requires those conditions. Descriptive complexity changes do not identify a population spectral-decay parameter. Across the representation comparison, fixed-kernel regularization paths, and stock-level attribution, the financial question is whether additional available directions improve the portfolio that finite historical information can support.